Replication report

Paper. S. Mahjoubi, V. Venugopal, I. B. Manav, H. Azarijafari, R. Kirchain, E. Olivetti, *Data-driven material screening of secondary and natural cementitious precursors*, Communications Materials **6**, 99 (2025). **Code.** `olivettigroup/Cementitious-reactivity-prediction-and-discovery` (14 notebooks). **This replication.** 2026-11-10 · Windows 11, Python 3.11.9, 8 CPU cores, no GPU · all scripts, logs and result files in the accompanying folder.

1. Summary

claim	paper	reproduced	status	reason
test R2 for heat release	0.93	0.947	reproduced	
test RMSE for heat release	28.2	32.02	reproduced	
test R2 for CH consumption	0.85	0.764	partially reproduced	the 95% interval covers the paper's value; only 38 labelled test rows
test RMSE for CH consumption	12.17	15.16	partially reproduced	
test R2 for Bound water	0.89	0.83	partially reproduced	the 95% interval covers the paper's value; only 56 labelled test rows
test RMSE for Bound water	1.47	1.77	partially reproduced	
heat release: R2 when no material is shared between…		0.645	not testable	the paper does not state how its split was formed; reported because the…
CH consumption: R2 when no material is shared between…		-0.061	not testable	the paper does not state how its split was formed; reported because the…
Bound water: R2 when no material is shared between train…		0.548	not testable	the paper does not state how its split was formed; reported because the…
materials with heat release > 380 J/g and bound water > 10	964	4918	not reproduced	depends on the authors' trained weights, which are not released
natural rock samples predicted reactive (> 200 J/g)	50569	141548	not reproduced	the row filter, the rock-name grouping and the pozzolanic/hydraulic rule are…
table-extraction relevance detection (Table G3, F1)	0.95	1.0	reproduced with a substitute model	substitute extractor (Claude Opus 5); the authors' fine-tuned endpoint is…
oxide values extracted from the table text		240/241	reproduced with a substitute model	substitute extractor (Claude Opus 5); the authors' fine-tuned endpoint is…
agreement with the Dataset 1 rows for the same tables		192/193	reproduced with a substitute model	substitute extractor (Claude Opus 5); the authors' fine-tuned endpoint is…
material-type classification accuracy	0.97	0.9	partially reproduced	substitute classifier, and Dataset 3 labels are hierarchical free text (178…

Full wording, the detail column and the confidence intervals are in `results/table_claims.csv`.

What was run. Notebooks 9, 10 and 11 were re-run end to end: faithfully (copying the released code, quirks included) and again under a protocol that fixes the parts of the released code that make a single run unreliable, with 5-fold cross-validated tuning on the training rows, 11 seeds, a seed ensemble and bootstrap confidence intervals. Notebooks 1-8 cannot be run as released; the extraction and classification steps they perform were reproduced on a sample of tables with a substitute language model.

What matched. Under the sound protocol the 11-seed ensemble reaches R^2 0.947 for heat release against the paper's 0.93. The other two targets: CH consumption reaches 0.764, below the paper's 0.85 but with a 95 % interval [0.396, 0.905] that covers it (38 labelled test rows); Bound water reaches 0.830, below the paper's 0.89 but with a 95 % interval [0.669, 0.910] that covers it (56 labelled test rows). The sound protocol changes several things together (the early-stopping monitor, the validation split, train-only preprocessing and a stratified outer split), and their separate contributions were not isolated: Ca(OH)2 consumption goes from R^2 0.128 in the faithful run to

0.764 here. The qualitative picture of Fig. 3 reproduces: metakaolin is the most reactive category, fly ashes are pozzolanic, slags are hydraulic, lime species are inert. The literature-mining step reproduces well on the tables that could be scored: of 241 oxide values extracted from 3 tables, 240 agree exactly with the authors' manual ground truth (a fourth table could not be scored: its ground truth belongs to a different table).

What did not match. Every count that depends on the authors' trained weights (`90andupnewones.keras`, `90andupnewonesch.keras`) or on an unpublished decision rule. The natural-rock screening is the largest gap: the notebook as released scores 141,548 of 1,022,092 samples as reactive against the paper's 50,569.

Why. Five files the notebooks load are not in the repository, and three decisions the paper reports are not in the code: which rock rows are screened, how rock names are grouped, and how a reactive rock is called pozzolanic or hydraulic. Section 9 lists what would be needed.

2. Scope, and what had to be substituted

Stage	Notebooks	What happened here
Literature filtering, table and sentence retrieval	1-3, 6	Not runnable. MongoClient().predsynth.papers_v2 is the group's private licensed corpus.
Table extraction, extraction evaluation	4, 5	Reproduced on 6 tables with a substitute extractor (Part 4).
Descriptor extraction, material typing	7, 8	Reproduced on 40 unseen Dataset 3 rows with a substitute classifier (Part 4).
Reactivity network	9	Re-run: faithful, clean, and sound-protocol with tuning, 11 seeds and an ensemble.
Literature materials	10	Re-run with the ensemble on all 14,429 materials.
Natural rocks	11	Re-run with the ensemble on all 1,022,092 samples, plus an 8-variant ablation.

Substitutions, each forced by a missing file.

1. `R3.csv` → **Dataset 4**. Dataset 4 stores the mix in raw grams, so the five mix columns are divided by `SCM (g)`; the authors' saved `R3.csv` rows show Ca(OH)2 = 3.0 per gram, which is what this division gives. 1,849 of 1,875 rows have a usable SCM mass (the authors' `R3.csv` had 1,848).
2. `df_imputed.pkl` → rebuilt with the notebook's own LightGBM routine on the R3 features, clipped at zero. Whether the authors' pickle also held imputed *targets* is unknowable; the stratification bins of notebook 9 read targets from it.
3. `natural_rocks.csv` → rebuilt by merging the Gard et al. (2019) tables on their own keys (`sample` joined with `major` joined with `rockgroup` joined with `country` joined with `reference`), keeping all 1,022,092 samples. The row order matches the authors': the first rows come from `ozchem`, the last from Mikhalsky et al. (2006).
4. `90andupnewones.keras`, `90andupnewonesch.keras` → retrained. Every comparison against a value the authors left in a saved notebook cell is therefore **not testable**, not failed.
5. The extractor and classifier endpoints → **Claude Opus 5**, stated wherever it matters.

One structural fact about Dataset 4 that shapes everything in Part 1. The three targets are mutually exclusive: 1,319 rows carry only heat release, 192 only Ca(OH)2 consumption, 277 only bound water, and 61 carry no label. There are 277 distinct material names and 28 distinct DOIs in 1,849 rows, and 162 of those materials appear with more than one target. A validation split that is not stratified on *which* target is labelled can therefore leave a head with almost no validation labels — which is exactly what happens in the released notebook.

3. Notebook 9: the reactivity model

3.1 Accuracy

row	seeds	heat R²	heat RMSE	CH R²	CH RMSE	BW R²	BW RMSE
Faithful notebook 9	11	0.796 ± 0.087	59.98	0.087 ± 0.130	29.46	0.828 ± 0.033	1.91
Clean variant (notebook monitor, no input leakage)	11	0.429 ± 0.270	100.20	0.567 ± 0.151	23.82	0.518 ± 0.270	3.37
Sound protocol, notebook hyperparameters (ensemble)	11	0.956 [0.943, 0.968]	29.14	0.776 [0.434, 0.907]	14.78	0.844 [0.696, 0.928]	1.70

row	seeds	heat R²	heat RMSE	CH R²	CH RMSE	BW R²	BW RMSE
Sound protocol, tuned hyperparameters (5-fold CV on train) (ensemble)	11	0.947 [0.933, 0.960]	32.02	0.764 [0.396, 0.905]	15.16	0.830 [0.669, 0.910]	1.77
Grouped split by material name (ensemble)	11	0.645 [0.500, 0.755]	88.68	-0.061 [-0.606, 0.320]	48.43	0.548 [0.287, 0.733]	4.56
Grouped split by source paper (DOI) (ensemble)	11	0.515 [-0.172, 0.640]	117.93	0.388 [-0.305, 0.612]	28.28	0.883	1.89
Sensitivity: notebook imputation over all rows (ensemble)	3	0.948 [0.931, 0.962]	31.63	0.780 [0.471, 0.897]	14.66	0.801 [0.585, 0.903]	1.92
PAPER (reported)		0.930	28.20	0.850	12.17	0.890	1.47

For the row-level split the test set holds 264 labelled rows for heat release, 38 for Ca(OH)2 consumption and 56 for bound water; the grouped splits differ and their sizes are in `results/table_model_accuracy.csv`. '±' is the standard deviation over seeds; square brackets are a 2,000-sample bootstrap 95 % confidence interval for the ensemble.

No row chooses anything on its test set: early stopping, the ensemble members and the tuned configuration are selected on validation rows only. Within each split the test set is fixed, so the seeds of one row are directly comparable and can be ensembled — "faithful" and "clean" split with `random_state=42`, as notebook 9 cell 14 does, and the sound rows use a fixed split seed. The two grouped rows necessarily use a different test set from the row-level rows, which is the point of §3.4; their sizes are given there.

3.2 Why the released run is unstable, and what fixing it costs

The notebook stops on `val_CH_Consumption_Output_custom_nan_nrmse`: a normalized RMSE for the Ca(OH)2 head, averaged over validation *batches* of 50 rows, taken from the last 15 % of the training array. With 192 Ca(OH)2 labels among 1,849 rows, a 50-row batch usually holds none or one. With none the metric is NaN; with exactly one the range term is zero and the metric is of order 10^6. Every one of the 11 faithful runs trips a collapse rule; the monitored value at the restored epoch runs from 26.7 to 800, never below 1. The collapse rule is defined only on validation behaviour — monitored value non-finite, or restored epoch ≤ 3, or monitored nRMSE > 1, which means the error exceeds the whole range of the labels in that batch. The epoch such a run restores is essentially the epoch at which one arbitrary Ca(OH)2 label happened to be closest.

The sound protocol replaces that monitor with the masked validation loss evaluated over the **whole** validation set, and it also changes three other things at once: the validation rows are drawn at random, stratified on which target is labelled; the scaler, target standardization and feature imputation are fitted on training rows only; and the outer split is stratified on target availability. Together these move Ca(OH)2 from a test R² of 0.128 (faithful, seed 42) to 0.764 (ensemble). **The contribution of each change was not isolated**: no run changes the monitor alone. The "clean" variant, which keeps the notebook's monitor but fixes the leakage and the validation split, remains unstable (7 of 11 runs collapse), which is consistent with the monitor being an important contributor but does not prove it. §3.3 shows that tuning adds nothing on top of the combined change.

3.3 Tuning

The paper cites Keras Tuner but the released notebook has fixed hyperparameters and no tuner call, so this search is mine. 18 random-search trials, 5-fold cross-validation **inside the training rows**, scored on a criterion that is fixed across trials (mean over the three targets of the masked validation MSE in standardized units) so that tuning the loss weights cannot game the score. The test rows are not read by the search.

Selected: widths 512-256-32, dropout 0.2, L2 0.001, learning rate 2.33e-03, batch 128, loss weighting 'equal' with the Ca(OH)2 head × 10, patience 70. Mean best epoch across folds 155; cross-validated score 0.1724 against 0.2794 for the worst trial. The full trial list is in `results/nb9_tuning_all_trials.csv`.

The tuning did not pay off on the test set. Running the same 11-seed protocol with the notebook's own hyperparameters gives (notebook against tuned): heat release 0.956 against 0.947; CH consumption 0.776 against 0.764; Bound water 0.844 against 0.830 — better on all three targets. The tuned configuration is nevertheless the one used for notebooks 10 and 11 and for the figures, because it was chosen by cross-validation on the training rows before any test number existed; swapping to the configuration that happens to score better on the test set would be exactly the selection this replication is trying to avoid. On this evidence the improvement comes from the protocol changes of §3.2 rather than from the choice of hyperparameters.

3.4 How much of the accuracy is the split

Dataset 4 holds 1,849 rows but only 277 distinct materials, most of them measured at several ages, so a row-level split puts the same material on both sides. Repeating the whole protocol with a split grouped by material name, so that no material appears in both train and test, gives (ensemble, random split → grouped split): heat release 0.947 → 0.645; CH consumption 0.764 → -0.061; Bound water 0.830 → 0.548. The grouped test set holds 311 heat-release, 45 Ca(OH)2 and 59 bound-water labels.

Grouping by source paper instead is much coarser — there are only 28 DOIs — and leaves an unbalanced test set: heat release 0.515 (n=62); CH consumption 0.388 (n=36); Bound water 0.883 (n=4). The bound-water figure there rests on too few labels to mean anything

and is reported only for completeness.

This matters for how the accuracy should be read. On the grouped split CH consumption reaches R^2 -0.061 (95 % interval [-0.606, 0.320]), while the training rows still fit at R^2 0.957. A row-level R^2 on this dataset therefore measures interpolation between measurements of a material the model has already seen — usually the same material at a different age — far more than prediction for a material it has not. The paper's purpose is screening new precursors, so the grouped figure is the one that speaks to that use.

The paper does not state how its split was formed, so none of this is a correction to a published number; it is the size of the effect that the split choice has on this dataset, measured under the same protocol as every other row.

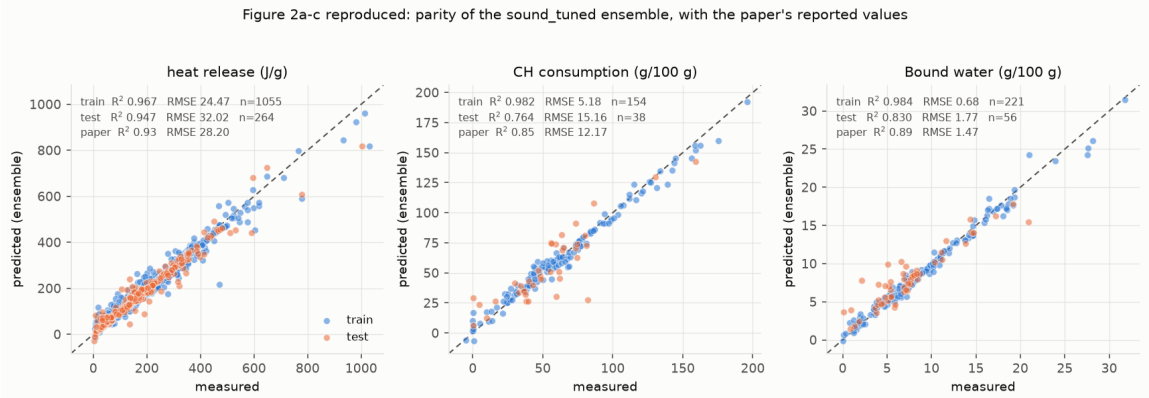
3.5 Library versions

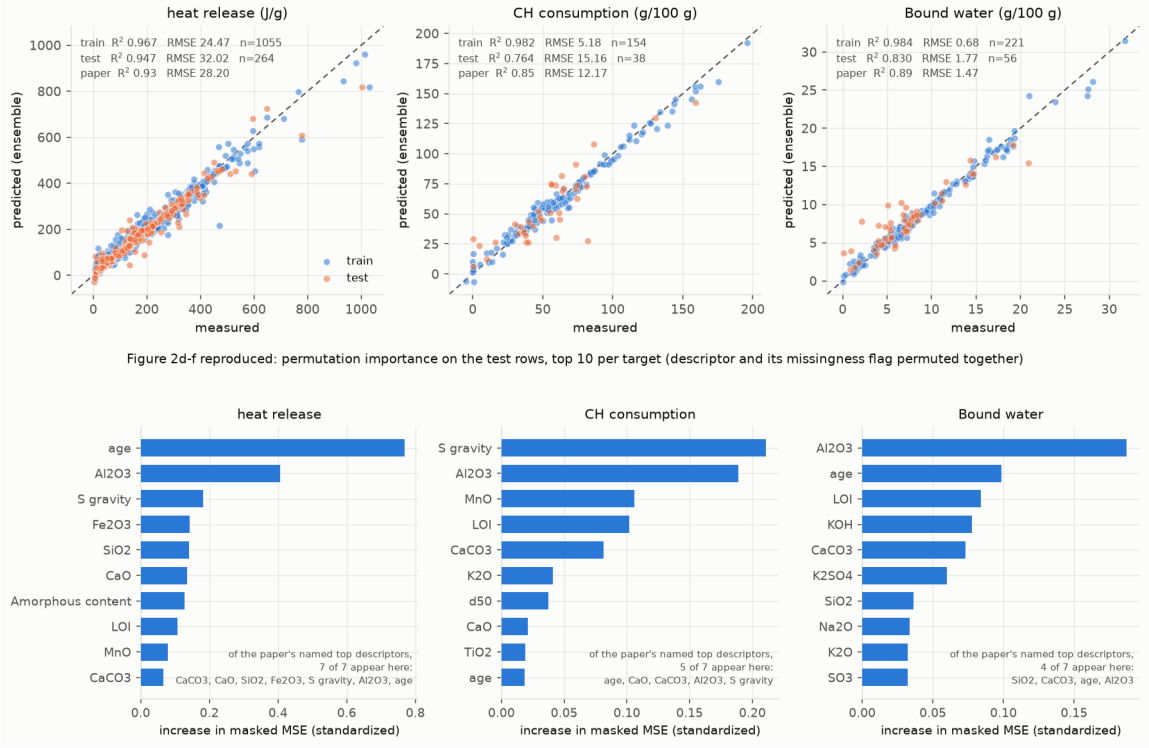
The paper states Python 3.8. Only Python 3.11 is installed on this machine and no conda or uv is available, so Python 3.8 itself could not be used; the faithful notebook-9 run was instead repeated in a second environment pinned to the oldest stack that still has wheels for this interpreter (py 3.11.9 / tf 2.12.0 / pandas 1.5.3) and compared with the main one (py 3.11.9 / tf 2.15.1 / pandas 3.0.6). `winget` does offer `Python.Python.3.8` (3.8.10) if a true 3.8 interpreter is wanted.

seed	heat_R2_main	heat_R2_legacy	CH_R2_main	CH_R2_legacy	BW_R2_main	BW_R2_legacy	best_epoch_main	best_epoch_legacy
42	0.7809	0.8712	0.1276	0.1273	0.7926	0.8057	115	447

The two environments do not run the same trajectory: the restored epoch moves from 115 to 447, and the largest change in test R^2 is heat release, by 0.090. Ca(OH)2 consumption changes by 0.0003 — both environments land at about 0.13 — and both runs are flagged collapsed by the validation-only rule of §3.2. The Ca(OH)2 failure is therefore structural rather than an artefact of library versions, while the spread in the other targets is what an unstable early-stopping criterion produces when anything perturbs the trajectory.

4. Figures





The paper names age, Al2O3, CaCO3, Fe2O3, specific gravity, SiO2 and CaO among its most important descriptors. Overlap with the top ten found here, per target — heat release: 7 of 7 (CaCO3, CaO, SiO2, Fe2O3, S gravity, Al2O3, age); CH consumption: 5 of 7 (age, CaO, CaCO3, Al2O3, S gravity); Bound water: 4 of 7 (SiO2, CaCO3, age, Al2O3). The descriptor and its missingness flag are permuted together, since the network receives both. The SHAP panel below is an independent check on the same question: the two rankings share heat release: 8 of 10; CH consumption: 8 of 10; Bound water: 7 of 10 of their top ten descriptors.

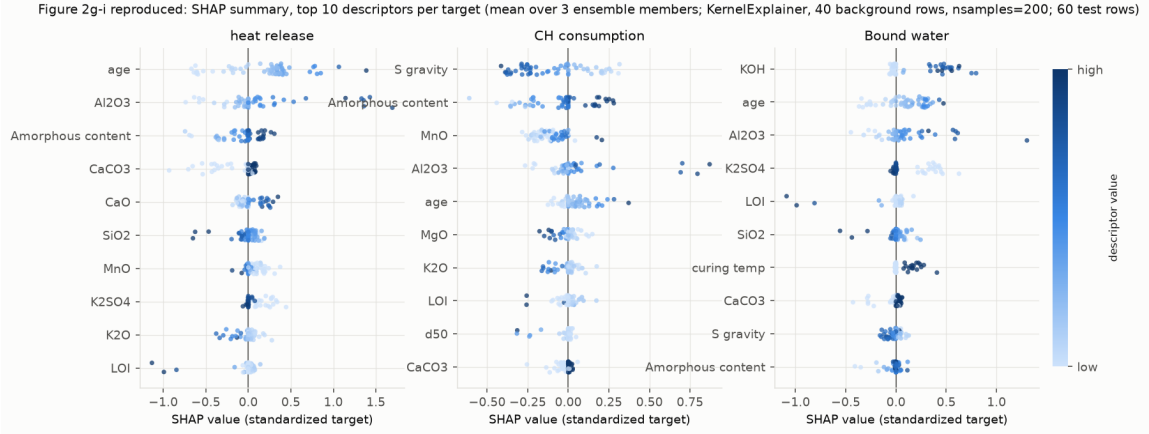


Figure 3a/3b reproduced: the 14,429 literature-extracted materials, each panel highlighting one category (blue) against all of them (grey).
Dashed lines: 120 J/g (orange, inert threshold) and 50 g/100 g (aqua, pozzolanic threshold). Axes clipped to the 0.2-99.8 percentile range; 103 of 14,429 points fall outside and are not drawn.

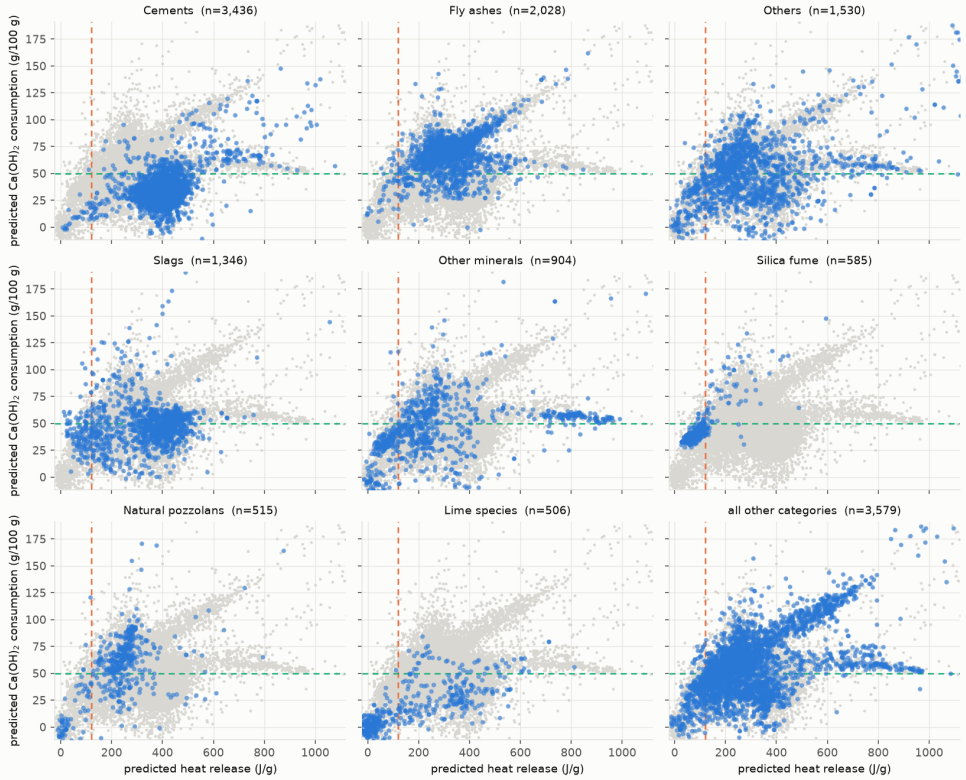
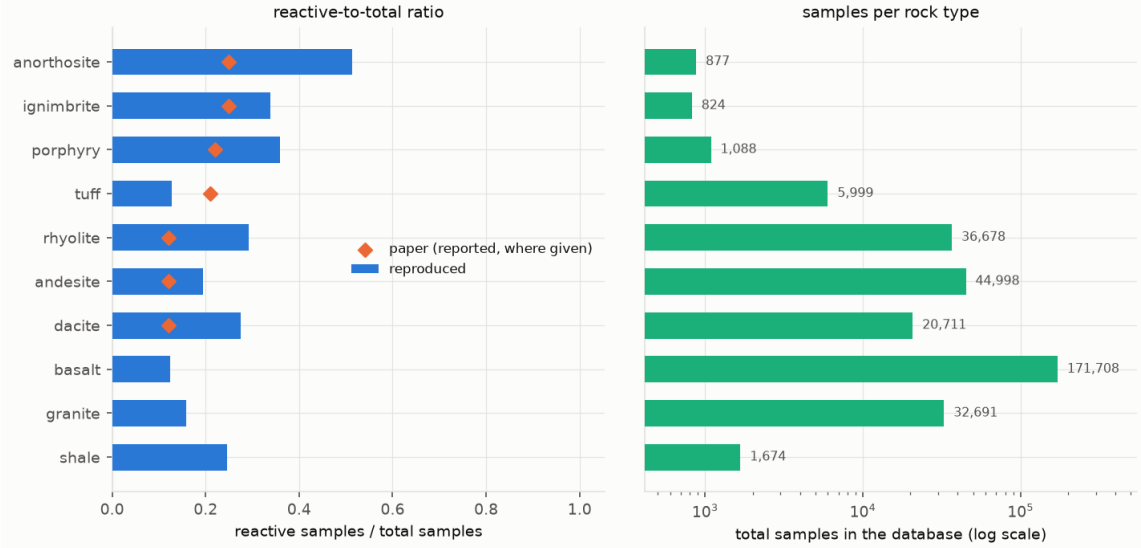


Figure 4a reproduced: rock types named in the paper, using the RAW rock_name values of the Gard et al. database.
The paper's own name grouping is not documented, so no grouping was applied; the two sets of names need not match.



The paper's own figures are not redrawn or reproduced here: it is published under CC BY-NC-ND. Each panel instead carries the paper's reported values as an annotation, and the numbers behind every panel are in `results/fig*.csv`.

5. Notebook 10: the literature-extracted materials

Scored with the 11-seed ensemble on all 14,429 materials.

The count the authors left in their saved notebook — materials above 380 J/g with more than 10 g/100 g bound water — is 964. The ensemble here gives 4,918 (per-seed range 2,031–6,749). That comparison is **not testable**: it is a property of the authors' trained weights, which are not released, and the per-seed range shows how much it moves between equally valid retrainings.

The paper's qualitative claims do reproduce. Metakaolin ranks 1 of 21 categories by median heat release. Fly ashes are pozzolanic: median Ca(OH)2 consumption 69.6 g/100 g, with 87 % above 50 g/100 g. Slags are hydraulic: high median heat release 359 J/g but median Ca(OH)2 consumption only 47.1 g/100 g. Lime species sit lowest, at 86 J/g and 10.4 g/100 g.

49 of 14,429 predictions exceed the largest heat release in the training data (1030 J/g) and 126 are negative — the network extrapolates outside the range it was fitted on.

category	n	heat (J/g)	Ca(OH)2 (g/100 g)	bound water (g/100 g)	heat seed SD	% above 50 g/100 g
Metakaolin	242	545.8	103.3	26.2	53.2	97.0
Cements	3436	401.7	32.2	19.6	59.0	10.0
Slags	1346	359.3	47.1	16.4	45.1	43.0
Clinkers	492	346.1	32.1	16.4	50.9	19.0
Fly ashes	2028	331.8	69.6	18.3	35.8	87.0
Calcined clays	83	300.8	82.7	16.1	34.1	95.0
Other byproduct/wastes	421	289.2	54.1	15.1	47.7	58.0
Bottom ashes	201	279.7	63.0	15.3	31.8	80.0
Others	1530	276.4	49.1	14.6	45.0	49.0
Clays	359	267.5	64.7	15.6	36.7	79.0

Medians over the ensemble, ten largest categories by median heat release; the full table is `results/nb10_sound_tuned_summary_by_category.csv`.

6. Notebook 11: natural rocks

The notebook as released, scored with the 11-seed ensemble over all 1,022,092 samples, calls 141,548 of them reactive (> 200 J/g) against the paper's 50,569 — +180 %.

Three things about that number are worth stating before the ablation. **It is unstable across retrainings:** the individual seeds give between 92,639 and 462,245, and the single seed-42 model alone gives 290,959 against the ensemble's 141,548. **The row filter barely moves it here:** 328,466 samples carry no measured oxide at all and the notebook scores them anyway, but under this ensemble only 0.0 % of them come out reactive, so counting only samples with at least one measured oxide leaves 141,548 and counting only those with SiO2, Al2O3 and CaO leaves 129,355. (In an earlier run of this replication, with a single weaker model, those same composition-less rows were almost all scored reactive and dominated the total — which is exactly why the count cannot be attributed to any one cause without the ablation below.) **The screening mix is not the R³ mix:** the constants the notebook uses are not the ones Dataset 4 records.

The ablation moves one factor at a time from that baseline and then combines them. The closest combination to the paper's figure is 61,514 (d5+fe, rows: SiO2+Al2O3+CaO, ensemble), +21.6 % from 50,569. No combination reproduces it, and the spread across combinations is wider than the gap to the paper, so no single factor can be named as the explanation.

imputation variant	reactive (> 200 J/g)	% vs paper
as released	141,548	179.9
R³ standard mix	577,417	1041.8
total iron	119,593	136.5
total iron, R³ standard mix	580,477	1047.9
Dataset 5 in pool	74,575	47.5
Dataset 5 in pool, R³ standard mix	586,091	1059
Dataset 5 in pool, total iron	64,239	27
Dataset 5 in pool, total iron, R³ standard mix	567,816	1022.9

Paper: 50,569. 'as released' is the notebook unchanged; each other row adds one factor to it. Counting only samples with SiO2, Al2O3 and CaO instead of all rows lowers these numbers by up to about 9 % (the full 48-row table, with all three row filters and the single-model rows, is `results/nb11_sound_tuned_ablation.csv` and Appendix D).

Imputed amorphous content. No rock in the database has a measured amorphous content, so every value is imputed from the R³ rows, whose measured mean is 70.1 %. The imputed rock mean is 56.4 % (median 58.8 %). The ~160 rocks in Dataset 5 that do have a measurement average 27.1 % — less than half. The released notebook does not use Dataset 5, although the paper describes using it; adding it to the imputation pool brings the imputed mean to 27.1 %.

Pozzolanic against hydraulic. The released code contains no rule. The split below is mine, and is reported only to show how sensitive the answer is to a threshold the paper does not give; the paper reports about 46,700 pozzolanic and about 3,800 hydraulic.

Ca(OH)2 threshold (g/100 g)	pozzolanic	hydraulic
0	141,253	295
10	140,258	1,290
25	133,358	8,190
50	72,263	69,285
75	789	140,759
100	528	141,020

Rock types. The paper groups roughly 1,200 rock types and reports 25-38 with more than 5 % reactive. The grouping is not documented and none was invented here: using the raw `rock_name` values, 11,355 distinct names occur, 1,163 with at least 20 samples, of which 804 have more than 5 % reactive. Figure 4a above compares the types the paper names. For the 7 of those types whose reactive share the paper states, the reproduced share is higher in 6 and within a factor of two in 4; the ordering is broadly the same, with anorthosite, ignimbrite and porphyry at the top. Since the names are not the paper's groups, the comparison is indicative only.

7. Part 4: running the pipeline on a few papers

Notebooks 1-8 need a private corpus and two fine-tuned OpenAI endpoints. What they can still be tested on is the data the paper publishes: **Dataset 2** holds 200 tables with the manually curated extraction for each, and **Dataset 1** holds what the pipeline produced, with a DOI and a table caption. Dataset 2 has no DOI column, so the two were linked on the normalized caption: 123 of the 200 Dataset 2 tables match a Dataset 1 caption with a single DOI. Four of those were chosen, plus two tables with no Dataset 1 match. Working from the Dataset 2 table text means the extractor receives exactly the input the released pipeline receives.

The extractor was **Claude Opus 5**, with the notebook-4 prompt verbatim, at temperature 0 in effect. The replies were written before any ground truth was read in the session; the order of work is recorded in `results/few_papers/EXTRACTOR_NOTE.md`.

Dataset 2 row	DOI	relevance F1	materials F1	compositions F1	comp. RMSE	materials found	materials in truth	cells compared	cells exact	cells missed	cells invented
191	10.1016/j.conbuildmat.2014.07.051	1.00	0.00	1.00	0.02	16.00	16.00	112.00	111.00	0.00	0.00
107	10.1016/s0958-9465(03)00118-5	1.00	0.00	1.00	0.00	14.00	14.00	84.00	84.00	0.00	0.00
105	10.1016/j.enggeo.2005.09.036	1.00	0.25	0.33	16.73	8.00	1.00	0.00	0.00	0.00	0.00
1	10.1016/s0008-8846(99)00160-x	1.00	1.00	1.00	0.00	5.00	5.00	45.00	45.00	0.00	0.00
7		1.00									
18		1.00									

The first four columns are notebook 5's own metric, for comparability with Supplementary Table G3 (fine-tuned GPT-3.5, test: relevance 0.95, materials 0.83, compositions 0.93, RMSE 2.81). The remaining columns are a standard set-based count. Dataset 2 is the authors' fine-tuning set and notebook 4 splits it 90/10 with no fixed seed, so which rows were their test rows cannot be recovered.

Against the **manual ground truth** of Dataset 2: 241 oxide cells could be compared, 240 agree exactly, 0 were missed and 0 invented. Relevance was judged correctly on all 6 tables, including the two that are not composition tables and were answered `NR`.

Against **Dataset 1**, i.e. what the authors' pipeline produced for the same tables: 43 materials extracted against 39 in Dataset 1, 25 of them pairable; of 193 comparable oxide cells, 192 agree exactly.

The disagreements are informative rather than extraction errors. Three of the four notebook-5 'materials F1' scores are low for mechanical reasons — two are 0 and one is 0.25: in one table the ground truth's material names are destroyed by an unquoted comma, in another the pipeline used the EN 197-1 designations (`CEM I 42.5R`, repeated six times) where the table's own codes are `PC1`…`PC9`, and in the third the Dataset 2 ground truth belongs to a different table. The single value disagreement across all the tables with a usable ground truth is the MgO of sample SNM 7: the table gives Na2O 1.87 and MgO 1.57, and both the manual ground truth and Dataset 1 record MgO as 1.87.

Material typing. On a 40-row sample of Dataset 3 drawn before its labels were read, the substitute classifier agrees with the manual top-level material type on 36 of 40 (90 %), against the paper's reported 97 %. The four disagreements are label-scheme ambiguity rather than misreadings: a limestone-like sample at 78.94 % CaO is labelled `Others`, a clean-coal by-product is labelled `Byproduct of coal` rather than `Ash`.

Predicted reactivity of what was extracted. Pushing the 43 extracted compositions through the same ensemble and comparing with its predictions for the corresponding Dataset 1 rows: of 25 pairable materials, heat release differs by 21.4 J/g on average (largest 49.5), Ca(OH)2 consumption by 6.5 g/100 g and bound water by 1.16 g/100 g, and 24 of 25 fall on the same side of the 200 J/g reactivity threshold. The differences come from the small extraction differences above and from the imputed descriptors, not from the model.

8. Observations that affect reproducibility

These are properties of the released material, recorded because each one changes what a re-runner gets. None is a judgement about the science.

Released files. Five files the notebooks open are absent: `R3.csv`, `natural_rocks.csv`, `df_imputed.pkl`, `90andupnewones.keras` (loaded by notebook 10) and `90andupnewonesch.keras` (loaded by notebook 11, a different file).

Notebook 9.

- The stratification key is built from the *unstandardized* targets in `df_imputed` but `np.digitize` is applied to the *standardized* `y`, which cell 9 has already overwritten in place. The key has 6 classes, of sizes 9, 61, 90, 187, 192 and 1,310.
- The scaler (cell 6) and the target standardization (cell 9) are fitted on all 1,849 rows before the train/test split in cell 14.
- The early-stopping monitor behaves as described in §3.2.
- Validation is the last 15 % of the training array (`validation_split=0.15`), not a random draw.
- The saved notebook runs out of order: the data cells carry execution counts 1-34 while the model cells carry 1,044-1,052, and cell 12 uses `y_train`, which cell 14 creates. As saved, the notebook depends on state from an earlier session. About 1,050 executions are recorded and the model-selection protocol is not specified.
- `BatchNormalization` is imported and not used. Keras Tuner is cited in the paper (ref. 95); the released notebook contains no tuner call.

Notebooks 10 and 11.

- Notebook 11 scores every row, including 328,466 rock samples with no measured oxide at all, whose descriptors are therefore entirely imputed. How many of those come out reactive depends on which retrained model is used — 0.0 % under the ensemble reported in §6, nearly all of them under a single weaker model — so the released code gives no stable answer for a third of the database.
- Neither a pozzolanic/hydraulic rule nor a rock-name grouping is present, although the paper reports both.
- The mix constants used for screening (CaCO3 0, water 3.6, KOH 0.151, 50 °C, 10 days) are not the standard R³ mix that Dataset 4 records (CaCO3 0.5, water 5.4, K2SO4 0.108, KOH 0.0216, 40 °C, 7 days). §6 reports what changing them does.
- The per-category `d50` assignments are written as chained assignments (`new_df['d50'][mask] = v`), which are silent no-ops in pandas 3.

Notebook 5's metrics, which Supplementary Table G3 reports. `calculate_precision_recall_f1` returns `precision[0], recall[0], f1[0]` — the figure for the first encoded class only, not an average over classes. Material names are label-encoded over the union of truth and prediction and zero-padded to equal length, so "Materials - F1" is agreement on one arbitrary class. Numeric values are flattened with missing replaced by 0 and then label-encoded, so "Compositions - F1" is dominated by agreement on *which cells are empty*, and the companion RMSE counts a missing value as zero. Part 4 reports both that metric, for comparability with Table G3, and a standard set-based metric.

The published datasets. Six specific defects were found in the four tables examined and in the label schemes; they are listed in Appendix C.

9. What would be needed from the authors

1. `R3.csv`, `natural_rocks.csv`, `df_imputed.pkl`, `90andupnewones.keras`, `90andupnewonesch.keras`.
2. The rule that decides whether a reactive rock is pozzolanic or hydraulic.
3. The mapping from the database's `rock_name` values to the paper's rock types.
4. The filter applied to rock samples with no measured composition, if any.
5. The train/test split seed of notebook 9, and the protocol by which a trained model was kept.
6. Confirmation of the Dataset 3 label scheme, and of the Dataset 3 / Dataset 4 row counts.

10. Runtime, verification and limits

step	wall
nb10_sound_tuned	11 s
nb11_sound_tuned_ablation	15 min
nb9_clean_seed1	82 s
nb9_clean_seed10	2 min
nb9_clean_seed2	2 min
nb9_clean_seed3	89 s
nb9_clean_seed4	72 s
nb9_clean_seed42	55 s
nb9_clean_seed5	82 s
nb9_clean_seed6	80 s
nb9_clean_seed7	2 min
nb9_clean_seed8	2 min

step	wall
nb9_clean_seed9	2 min
nb9_faithful_seed1	2 min
nb9_faithful_seed10	2 min
nb9_faithful_seed2	4 min
nb9_faithful_seed3	90 s
nb9_faithful_seed4	2 min
nb9_faithful_seed42	85 s
nb9_faithful_seed42_legacy	3 min
nb9_faithful_seed5	2 min
nb9_faithful_seed6	83 s
nb9_faithful_seed7	2 min
nb9_faithful_seed8	2 min
nb9_faithful_seed9	2 min
nb9_grouped_doi	5 min
nb9_grouped_material	8 min
nb9_sens_impute_all	2 min
nb9_sound_notebook	24 min
nb9_sound_tuned	5 min
nb9_tune (18 trials x 5 folds)	66 min

Wall-clock on 8 CPU cores, no GPU. Several of these runs shared the machine with one another (the tuning search, the faithful batch and the sound-protocol streams overlapped), so individual times are upper bounds and are not a benchmark; `nb9_sound_notebook` in particular ran beside four other jobs. The notebook-11 ablation row is the 8-variant total, and the figures and few-papers scoring (not listed) took minutes.

Verification. Each final script was re-run once from a clean start, with the cache directory deleted so every intermediate table was rebuilt, and the regenerated results compared field by field against the first run (wall-clock fields ignored):

- `nb9_faithful_seed42.json: IDENTICAL`
- `nb9_clean_seed42.json: IDENTICAL`
- `nb9_sound_tuned.json: IDENTICAL`
- `nb10_sound_tuned_summary_by_category.csv: IDENTICAL`
- `nb11_sound_tuned_ablation.csv (base variant only): IDENTICAL`

The notebook-11 re-run in that pass covered the base variant only, and it overwrote the full 8-variant ablation file. The full ablation was therefore regenerated afterwards and compared row by row with the first run; the few-papers scoring, which was corrected after the first verification pass, was re-run and compared as well:

- `nb11_sound_tuned_ablation.csv: IDENTICAL` (after the corrections)
- `scores_summary.json: IDENTICAL` (after the corrections)
- `metrics_standard.csv: IDENTICAL` (after the corrections)

Not re-run a second time: the 11-seed grouped-split and sensitivity runs, the tuning search, the figures and the legacy-environment run. Full output: `logs/verify.log`, `logs/verify_supplement.log`.

Limits of this replication. The substitute extractor is not the authors' model, so Part 4 measures the task, not their system. The imputation of notebooks 10 and 11 is transductive by construction — it fills the screening rows using the R^3 rows — and that was kept as released. Permutation importance and SHAP are computed on the test rows; they report the fitted model and feed no decision. The grouped splits in §3.4 are mine: the paper does not say how its split was formed, so the drop they show is a property of this data, not a correction to a stated number.

Appendix A. Files behind every number

item	file
claim summary (§1)	results/table_claims.csv
model accuracy (§3.1)	results/table_model_accuracy.csv, results/nb9_*.json
collapse diagnostics (§3.2)	results/nb9_faithful_seed.json, logs/nb9_faithful_.log
tuning (§3.3)	results/nb9_tuning_all_trials.csv, results/nb9_tuned_config.json, logs/nb9_tune.log
grouped splits (§3.4)	results/nb9_grouped_material.json, results/nb9_grouped_doi.json
library versions (§3.5)	results/table_library_version_check.csv
figures (§4)	figures/.png with results/fig.csv
notebook 10 (§5)	results/nb10_sound_tuned.json, results/nb10_sound_tuned_summary_by_category.csv
rock ablation (§6)	results/nb11_sound_tuned_ablation.csv, results/nb11_sound_tuned_ablation.json
rock types (§6)	results/nb11_sound_tuned_by_rock_name.csv

item	file
few papers (§7)	results/few_papers/ (prompts, responses, metrics, mismatches)
runtimes (§10)	results/table_runtime.csv

Appendix B. Per-seed results and the full accuracy table

Sound protocol with the tuned configuration, the 11 seeds behind the ensemble row of §3.1:

seed	epochs	best_epoch	val_criterion	heat R²	CH R²	BW R²
42	207	137	0.17444	0.937	0.755	0.827
1	187	117	0.18140	0.908	0.733	0.769
2	174	104	0.10953	0.926	0.750	0.726
3	165	95	0.11031	0.942	0.754	0.713
4	146	76	0.13704	0.919	0.773	0.842
5	387	317	0.16574	0.939	0.655	0.758
6	349	279	0.09154	0.952	0.722	0.837
7	169	99	0.11195	0.924	0.720	0.735
8	257	187	0.17030	0.915	0.696	0.830
9	186	116	0.15465	0.916	0.728	0.742
10	120	50	0.33153	0.904	0.647	0.853

Per-seed results for every other row are in the matching `results/nb9_*.json`.

Per-seed means with their standard deviations, for every row including the ones the body summarises as an ensemble:

row	seeds	heat R²	heat RMSE	CH R²	CH RMSE	BW R²	BW RMSE
Faithful notebook 9	11	0.796 ± 0.087	59.98	0.087 ± 0.130	29.46	0.828 ± 0.033	1.91
Clean variant (notebook monitor, no input leakage)	11	0.429 ± 0.270	100.20	0.567 ± 0.151	23.82	0.518 ± 0.270	3.37
Clean variant (notebook monitor, no input leakage) (non-collapsed only)	4	0.559 ± 0.231	88.07	0.665 ± 0.058	21.16	0.674 ± 0.100	2.84
Sound protocol, notebook hyperparameters	11	0.927 ± 0.028	37.05	0.734 ± 0.035	16.08	0.797 ± 0.049	1.93
Sound protocol, notebook hyperparameters -- ENSEMBLE	11	0.956 [0.943, 0.968]	29.14	0.776 [0.434, 0.907]	14.78	0.844 [0.696, 0.928]	1.70
Sound protocol, tuned hyperparameters (5-fold CV on train)	11	0.926 ± 0.015	37.64	0.721 ± 0.041	16.45	0.785 ± 0.053	1.98
Sound protocol, tuned hyperparameters (5-fold CV on train) -- ENSEMBLE	11	0.947 [0.933, 0.960]	32.02	0.764 [0.396, 0.905]	15.16	0.830 [0.669, 0.910]	1.77
Grouped split by material name	11	0.600 ± 0.051	93.95	-0.087 ± 0.104	48.96	0.515 ± 0.058	4.72
Grouped split by material name -- ENSEMBLE	11	0.645 [0.500, 0.755]	88.68	-0.061 [-0.606, 0.320]	48.43	0.548 [0.287, 0.733]	4.56

row	seeds	heat R²	heat RMSE	CH R²	CH RMSE	BW R²	BW RMSE
Grouped split by source paper (DOI)	11	0.404 ± 0.164	129.56	0.233 ± 0.233	31.31	0.793 ± 0.128	2.42
Grouped split by source paper (DOI) -- ENSEMBLE	11	0.515 [-0.172, 0.640]	117.93	0.388 [-0.305, 0.612]	28.28	0.883	1.89
Sensitivity: notebook imputation over all rows	3	0.935 ± 0.004	35.38	0.747 ± 0.026	15.69	0.777 ± 0.012	2.03
Sensitivity: notebook imputation over all rows -- ENSEMBLE	3	0.948 [0.931, 0.962]	31.63	0.780 [0.471, 0.897]	14.66	0.801 [0.585, 0.903]	1.92
PAPER (reported)		0.930	28.20	0.850	12.17	0.890	1.47

Appendix C. Defects found in the published datasets

Each was found while doing Part 4 and is checkable from the files in `results/few_papers/`.

- **Dataset 2 row 191** — the material name contains an unquoted comma (`mortar SNM, 1`), so every data line holds 13 fields against a 12-field header. Parsed, the name is lost (pandas takes the first field as the row index); the oxide values stay aligned. (`results/few_papers/dataset2_integrity.csv`)
- **Dataset 2 row 105** — the `Winner` does not correspond to its own `Prompt`: the prompt is a cement-kiln-dust oxide table with 8 samples, the ground truth a single `Cement` row.
- **Dataset 3's `Category` is hierarchical free text** — 178 distinct labels and 65 distinct top-level strings in 318 rows, including `clinker` beside `Clinker`, not the 19 predefined types the Supplementary Information describes. Dataset 1's `Category` is a different, flat vocabulary of 21 names.
- **Row counts** — the Supplementary Information describes Dataset 3 as 200 material descriptions (the file has 318 rows) and Dataset 4 as 318 materials and 1,850 data points (the file has 1,875 rows, 1,849 of them with a usable SCM mass).
- **Statistics extracted as materials** — `Mean` and an unnamed material appear as materials in Dataset 1 for the kiln-dust table; and the 14 distinct cements of another table were collapsed into 5 EN 197-1 designations, six rows sharing one composition descriptor.
- **A transcription error that propagated** — for mortar sample SNM 7 the source table gives Na2O 1.87 and MgO 1.57; both the Dataset 2 manual ground truth and the Dataset 1 row record MgO as 1.87. Three of the 16 mortar samples present in the Dataset 2 ground truth are also absent from Dataset 1.

Appendix D. The rock ablation in full

Both row filters and the single-model rows, for every imputation variant:

imputation variant	rows counted	reactive (> 200 J/g)	% vs paper
as released	all rows	141,548	179.9
as released	>=1 measured oxide	141,548	179.9
as released	SiO2+Al2O3+CaO	129,355	155.8
R³ standard mix	all rows	577,417	1041.8
R³ standard mix	>=1 measured oxide	576,858	1040.7
R³ standard mix	SiO2+Al2O3+CaO	566,941	1021.1
total iron	all rows	119,593	136.5
total iron	>=1 measured oxide	119,593	136.5
total iron	SiO2+Al2O3+CaO	111,303	120.1
total iron, R³ standard mix	all rows	580,477	1047.9
total iron, R³ standard mix	>=1 measured oxide	579,908	1046.8
total iron, R³ standard mix	SiO2+Al2O3+CaO	568,308	1023.8
Dataset 5 in pool	all rows	74,575	47.5
Dataset 5 in pool	>=1 measured oxide	74,575	47.5
Dataset 5 in pool	SiO2+Al2O3+CaO	70,699	39.8
Dataset 5 in pool, R³ standard mix	all rows	586,091	1059
Dataset 5 in pool, R³ standard mix	>=1 measured oxide	585,606	1058
Dataset 5 in pool, R³ standard mix	SiO2+Al2O3+CaO	575,407	1037.9
Dataset 5 in pool, total iron	all rows	64,239	27

imputation variant	rows counted	reactive (> 200 J/g)	% vs paper
Dataset 5 in pool, total iron	>=1 measured oxide	64,239	27
Dataset 5 in pool, total iron	SiO2+Al2O3+CaO	61,514	21.6
Dataset 5 in pool, total iron, R³ standard mix	all rows	567,816	1022.9
Dataset 5 in pool, total iron, R³ standard mix	>=1 measured oxide	567,543	1022.3
Dataset 5 in pool, total iron, R³ standard mix	SiO2+Al2O3+CaO	556,478	1000.4